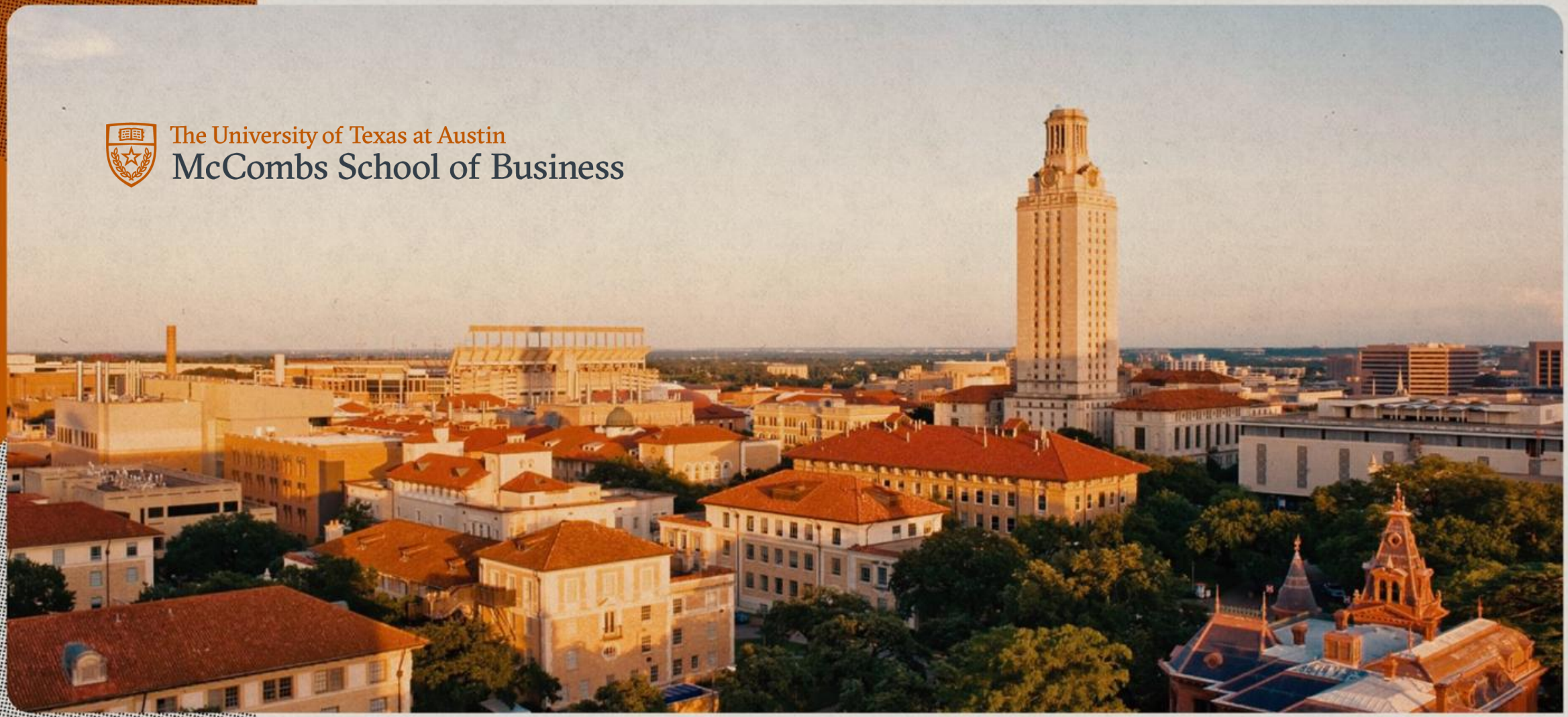

D1: Intro to Unsupervised Learning

Learning from Data

Tool Sets

Statistical Learning

- simpler algorithms
- better understood

Machine Learning

- more complex algorithms
- less well understood

SL = ML

Supervised Learning	Unsupervised Learning
$X \longrightarrow Y$	X
predict Y from X	discover structure of X
dependent variable	no dependent variable
training data	no training data

Supervised vs Unsupervised

In supervised data analytics there is a dependent variable.

In unsupervised data analytics there is no dependent variable.

In supervised data analytics there is a training dataset.

In unsupervised data analytics there is no training dataset.

Supervised or Unsupervised?

1. You want to find out what factors lead consumers to subscribe to your magazine.
2. You want to find out what demographic segments your subscribers fall into.
3. You want to find out how your profits depend upon your advertising expenditures.
4. You want to compress your customer data into fewer variables that are nearly as meaningful.
5. You want to forecast your profits for next quarter.
6. You want to uncover hidden attitudes in the American electorate.

Unsupervised Learning

$$X_{n \times p}$$

n rows of cases, records, observations, profiles

p columns of variables, fields, features

<u>Goal</u>: Discover row and column structure of X

Two Methodology Classes

Cluster Analysis

simplify X by grouping rows of X into clusters of similar rows

Dimensionality Reduction

simplify X by combining columns of X into new variables

Cluster Analysis Methods

K-means

Hierarchical agglomerative/divisive

GMM (Gaussian Mixture Models)

DBSCAN (Density-Based Spatial Clustering of Applications with Noise)

Association rule mining (Apriori)

(Extended) Isolation Forest method

One-Class SVM (Support Vector Machine)

SOM (Self-Organizing Map)

Dimensionality Reduction Methods

Principal Components Analysis

Factor Analysis

Multidimensional Scaling

Independent Component Analysis (ICA)

t-SNE (t-Distributed Stochastic Neighborhood Embedding)

Autoencoders

Related Applications: Association Rules

$X_{n \text{ rows}}$

Each row is an *itemset*. Number of *items* (columns) varies among rows.

(Ex: customer shopping baskets)

<u>Goal</u>: Learn which items tend to occur together and which do not.

Apriori algorithm

FP-Growth algorithm (Frequent Pattern Growth)

Related Applications: Anomaly Detection

$$X_{n \times p}$$

Goal: Learn which rows are anomalous (very unlike the other rows)

- [Extended] Isolation Forest method
- One-Class SVM (Support Vector Machine)
- Statistical methods

Advisory

This presentation is over-simplified.

There are many more unsupervised methods than listed herein.

Most methods were proposed b/c they sound good intuitively. The theoretical foundations may be weak. Justification is often empirical or by simulations.

Work goes on to find better methods and justify them.